Attribute GRR Study - Complete Guide

Kappa coefficient method for Go/No-Go and visual inspection gauges

QUALITY HEAD AGENTS | AIAG MSA 4th Edition | IATF 16949:2016

When to Use Attribute GRR

- Use Attribute GRR when your gauge or inspection method gives a PASS/FAIL result only.
- Examples: Go/No-Go gauges, torque stall, leak test (pass/fail), visual inspection, snap gauges.
- Cannot use variable GRR because there is no numerical measurement to calculate variation.
- AIAG MSA 4th Edition Chapter 4 covers Attribute studies.
- Kappa coefficient measures agreement between and within appraisers.

Study Method - 50 Parts, 3 Operators, 2 Trials

Part Selection:	Select 50 parts: approx. 20 clearly conforming, 20 clearly non-conforming, 10 borderline.
Reference Standard:	Establish true reference value using CMM or expert assessment. This is the 'standard' all results are compared to.
Blind Study:	Each operator evaluates all 50 parts twice. Operators must not know other results or previous own results.
Recording:	Record Pass/Fail for each operator, each trial, for each part. Compare to reference standard.
Randomization:	Randomize part presentation order between operators to prevent memory effects.

Kappa Coefficient - Calculation and Interpretation

Kappa Formula:	K = (Po - Pe) / (1 - Pe), where Po = observed agreement, Pe = expected agreement by chance
Kappa >= 0.75:	Acceptable - Good agreement between appraisers
Kappa 0.50-0.74:	Marginal - May be acceptable for non-critical characteristics. Requires improvement plan.
Kappa < 0.50:	Unacceptable - Gauge/inspection method cannot reliably distinguish conforming from non-conforming.
Within Appraiser:	Each appraiser's two trials must also show Kappa >= 0.75. Low within-appraiser Kappa = repeatability problem.

Improving a Failed Attribute GRR

- Identify the borderline parts causing disagreement - these reveal the acceptance boundary problem.
- Define clear visual standard - create limit samples (golden samples) for the borderline condition.
- Develop Attribute Agreement Guide with photos of Pass, Fail, and Borderline conditions.
- Train all inspectors using the visual standard before re-study.
- Re-study after improvement. If Kappa still < 0.75, the inspection method is fundamentally unreliable.
- Consider upgrading from attribute to variable measurement (e.g., surface roughness tester vs visual).